

Small-angle scattering of interacting particles. II.[†] Generalized indirect Fourier transformation under consideration of the effective structure factor for polydisperse systems

B. WEYERICH, J. BRUNNER-POPELA AND O. GLATTER*

Physical Chemistry, University of Graz, Heinrichstraße 28, A - 8010 Graz, Austria.

E-mail: otto.glatte@kfunigraz.ac.at

(Received 30 April 1998; accepted 15 September 1998)

Abstract

The indirect Fourier transformation (IFT) is the method of choice for the model-free evaluation of small-angle scattering data. Unfortunately, this technique is only useful for dilute solutions because, for higher concentrations, particle interactions can no longer be neglected. Thus an advanced technique was developed as a generalized version, the so-called generalized indirect Fourier transformation (GIFT). It is based on the simultaneous determination of the form factor, representing the intraparticle contributions, and the structure factor, describing the interparticle contributions. The former can be determined absolutely free from model assumptions, whereas the latter has to be calculated according to an adequate model. In this paper, various models for the structure factor are compared, *e.g.* the effective structure factor for polydisperse hard spheres, the averaged structure factor, the local monodisperse approximation and the decoupling approximation. Furthermore, the structure factor for polydisperse rod-like particles is presented. As the model-free evaluation of small-angle scattering data is an essential point of the GIFT technique, the use of a structure factor without any influence of the form amplitude is advisable, at least during the first evaluation procedure. Therefore, a series of simulations are performed to check the possibility of the representation of various structure factors (such as the effective structure factor for hard spheres or the structure factor for rod-like particles) by the less exact but much simpler averaged structure factor. In all the observed cases, it was possible to recover the exact form factor with a free determined parameter set for the structure factor. The resulting parameters of the averaged structure factor have to be understood as apparent model parameters and therefore have only limited physical relevance. Thus the GIFT represents a technique for the model independent evaluation of scattering data with a minimum of *a priori* information.

1. Introduction

The interpretation of small-angle X-ray and neutron scattering data plays an essential role in the determination of size, shape and internal structure of colloidal particles in the size range from a few nanometers up to about a hundred nanometers (Guinier & Fournet, 1955; Glatter & Kratky, 1982; Feigin & Svergun, 1987; Brumberger, 1995). As long as the solutions are dilute, there exists a direct relation between the particle [represented by its pair-distance-distribution function $p(r)$] and its scattering data by means of a Fourier transformation. If the concentrations increase, however, in addition to intraparticle scattering, interparticle scattering must be taken into consideration (Ailawadi, 1980; Hayter & Penfold, 1981; Hayter, 1985; Kaler, 1988, 1995; Hansen & McDonald, 1990; D'Aguanno & Klein, 1996). The former can be described by the form factor $P(q)$, and the latter by the structure factor $S(q)$, where q is the length of the scattering vector.

We have developed a technique to determine both the form factor and the structure factor simultaneously. It can be regarded as a generalization of the well known indirect Fourier transformation (IFT) (Glatter, 1977*a,b*; 1980*a,b*) and therefore we call it generalized indirect Fourier transformation (GIFT). The basic principles were published in the first part of this series (Brunner-Popela & Glatter, 1997). The form factor $P(q)$ is determined absolutely model free, whereas the structure factor $S(q)$ has to be calculated according to an adequate model. In the previous publication we restricted ourselves to a very simple model for $S(q)$, the so-called averaged structure factor. In this paper we want to extend our technique with regard to interaction effects, so we introduce the effective structure factor (Vrij, 1979; Blum & Stell, 1979) for hard spheres, taking polydispersity into account properly. Unlike the structure factor for monodisperse particles, this effective structure factor depends on both the particle interactions and the scattering properties of the single particles. This structure factor is compared to various approximations, like the local monodisperse approximation (Pedersen, 1994) or the decoupling approximation (Kotlarchyk & Chen, 1983). There exist pronounced differences in the shapes of the structure factors, espe-

[†] For Part I, *Basic Principles of a Global Evaluation Technique*, see Brunner-Popela & Glatter [*J. Appl. Cryst.* (1997), **30**, 431–442].

cially in the case of inhomogeneous particles with low mean contrast. Furthermore, the structure factor for rod-like particles including length polydispersity is presented. All these structure factors do not take charge into account. The method itself is able to implement electrostatic interaction *via* an adequate structure factor. Such an implementation is part of our ongoing research.

The data evaluation procedure of the GIFT is the solution of a highly nonlinear least-squares problem. In order to determine the form factor and the structure factor simultaneously, these two terms are represented by separate sets of unknown coefficients, which are calculated using an iterative sequential algorithm in which one set is kept constant while the other one is improved and *vice versa*. This procedure allows the use of an arbitrary model for the structure factor. The use of the effective structure factor in the data evaluation procedure, however, leads to numerical instabilities so that the procedure has to be modified in order to facilitate the determination of the global minimum of the parameter surface. Thus we have to deal with two coupled ill-posed problems, once in regard to the form factor and once to the structure factor.

The model-free evaluation of small-angle scattering data is a characteristic of our technique. Therefore, any structure factor including a form amplitude should be avoided to ensure model-independent evaluation if the scattering data are evaluated in a first run. For this reason, we have performed a series of simulations on the evaluation of scattering data using a different structure factor for the simulation and for the evaluation. We present results for the effective structure factor for hard spheres and for the structure factor for rod-like particles, both represented by the averaged structure factor. In both cases it is possible to recover the form factor correctly and to represent the simulated structure factors reasonably well by the simple averaged structure-factor model. The resulting parameters of this structure factor have to be understood as apparent model parameters with no or quite limited physical relevance. Nevertheless, we want to stress the point that during the evaluation of experimental data, the use of the averaged structure factor allows for a first determination of the particle size and shape. In a next step, this simple but general approximation for interparticle contributions can be replaced by a more realistic one depending on the structural information about the system determined in the first step. One of the most important types of structure-factor models is the effective structure factor for polydisperse spheres because it shows a very pronounced influence in the case of zero contrast.

2. Theory

2.1. Generalized indirect Fourier transformation

The evaluation of small-angle scattering data using the indirect Fourier transformation (Glatter, 1977a,b;

1980a,b) leads to the determination of the pair-distance-distribution function $p(r)$ containing information on size, shape and inner structure (Glatter, 1979). As long as concentrations are low enough so that interparticle interactions are negligible, there will not appear any problem during this procedure (see Figs. 1a and 1b). If concentrations increase, however, the scattering intensity $I(q)$ deviates from the ideal particle form factor $P(q)$ because of the interactions between the particles that are represented by the structure factor $S(q)$. Transforming $I(q)$ into real space without separating out the influence of $S(q)$ leads to a Fourier transform $i(r)$ that can hardly be interpreted because of strong oscillations at high r values. This total distribution function $i(r)$ describes the distance distributions of the overall density fluctuations around the mean value, which are related to the particle density distributions, described by the correlation function $\gamma(r)$, and to the total correlation

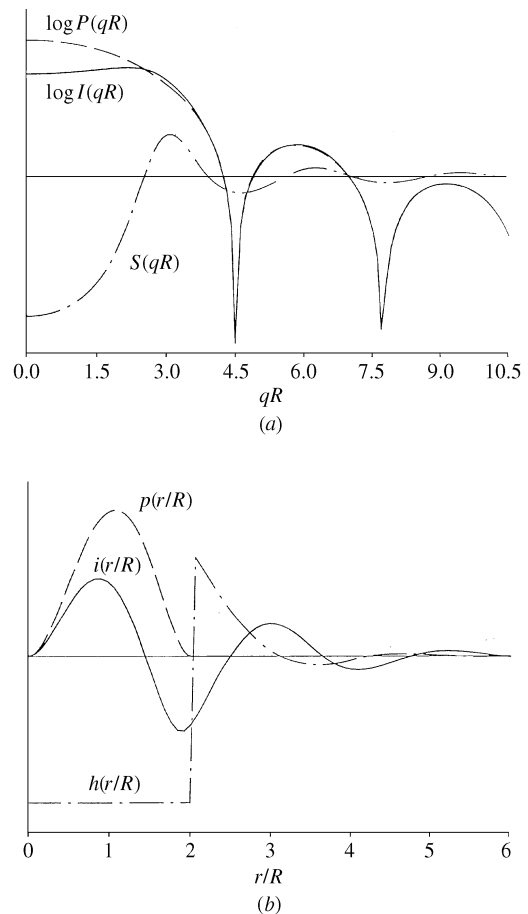


Fig. 1. (a) Form factor $P(q)$ (dashed line), structure factor $S(q)$ (dashed-dotted line), and scattering intensity $I(q)$ (full line) for monodisperse hard spheres. (b) Pair-distance-distribution function $p(r)$ (dashed line), total correlation function $h(r)$ (dashed-dotted line), and pair-distance-distribution function $i(r)$ resulting from $I(q)$ for monodisperse hard spheres (full line).

function $h(r) = [g(r) - 1]$, describing the deviation of the particle distribution from the statistical uncorrelated situation (D'Aguanno & Klein, 1996). The correct $p(r)$ can only be recovered under consideration and separation of the particle interactions during the evaluation procedure. We have developed a technique for a simultaneous determination of both the form factor $P(q)$, describing the *intraparticle* scattering contributions, and the structure factor $S(q)$, corresponding to the *interparticle* contributions (Brunner-Popela & Glatter, 1997). This technique can be understood as a generalized indirect Fourier transformation (GIFT).

In the case of a monodisperse, homogeneous and isotropic dispersion of spherical particles, no orientation effects must be taken into account so that the total scattering intensity $I(q)$ can be written as a product:

$$I(q) = NP(q)S(q). \quad (1)$$

Such a formal factorization is also possible for non-spherical or polydisperse systems. In these cases the structure factor has to be replaced by a so-called effective structure factor (see §2.2).

The technical details of the GIFT method were described in full detail in part I of this series (Brunner-Popela & Glatter, 1997) and are therefore not repeated here. In this procedure the structure factor $S(q)$ is determined by means of a nonlinear least-squares procedure including linearization of the fit function

$$I^{\text{exp}}(q) - \tilde{I}(q, \mathbf{d}) = \sum_{k=1}^4 \frac{\partial I(q, \mathbf{d})}{\partial d_k} (d_k^{(n+1)} - d_k^{(n)}) \quad (2)$$

with $I^{\text{exp}}(q)$ being the experimental data, $\tilde{I}(q, \mathbf{d})$ the approximation function and d_k the parameters of the structure factor. This leads to a set of normal equations that can be solved by inversion

$$\mathbf{t} = \mathbf{T}\Delta\mathbf{d}^{(n)} \quad \text{and} \quad \Delta\mathbf{d}^{(n)} = \mathbf{T}^{-1}\mathbf{t} \quad (3)$$

where $\mathbf{t}$ corresponds to the expansion coefficients (projection of the measured function on the basic functions) and $\Delta\mathbf{d}$ to the corrections for the solution coefficients (description of the structure factor). $\mathbf{T}$ is called the overlap matrix of the basic functions. In general there are no stability problems in this part of the calculation except for the effective structure factor, which will be described below in more detail.

Thus, both the form factor and the structure factor are determined simultaneously within one procedure. While the former is absolutely model free (just rough information on the maximum dimension of the particles is needed), the latter is calculated according to the Percus-Yevick approximation for hard spheres (Percus & Yevick, 1958; Percus, 1962) and therefore is not model independent. As we are dealing here with uncharged systems only, the model parameters are the volume fraction, radius and polydispersity of the spheres.

2.2. Structure factor for polydisperse systems

2.2.1. Spherical particles. This section is devoted to the description of interparticle contributions. As described above, interactions cause particle correlations that affect the scattering curve, especially at low angles. In mathematical terms, this effect can be described by the product of the form factor and the structure factor [equation (1)], where the latter is defined as the thermodynamic ensemble average over the Fourier components of the particle density fluctuations. For the special case of monodisperse spherical particles, the interparticle contributions are completely decoupled from the scattering properties of a single particle. Therefore, the structure factor depends only on the correlations caused by the intermolecular interaction potential.

For an m -component mixture of spherical particles, the expression for the scattered intensity becomes

$$I(q) = N \sum_{\alpha, \beta=1}^m f_{\alpha} f_{\beta} B_{\alpha}(q) B_{\beta}(q) S_{\alpha\beta}(q) \quad (4)$$

where f_{α} is the form amplitude of species α at $q=0$, $B_{\alpha}(q)$ the normalized form amplitude of species α , and $S_{\alpha\beta}(q)$ the partial structure factor. In order to obtain an expression for the scattered intensity which is similar to the monodisperse case as presented in (1), (4) is rewritten in the following form:

$$I(q) = N \overline{P(q)} S^{\text{eff}}(q). \quad (5)$$

The averaged form factor $\overline{P(q)}$, with x_{α} being the molar fraction of species α , is given by

$$\overline{P(q)} = \sum_{\alpha=1}^m x_{\alpha} f_{\alpha}^2 B_{\alpha}^2(q) \quad (6)$$

and the 'effective' structure factor is defined by (Pusey *et al.*, 1982; Salgi *et al.*, 1993)

$$S^{\text{eff}}(q) = \frac{1}{\overline{P(q)}} \sum_{\alpha, \beta=1}^m f_{\alpha} f_{\beta} B_{\alpha}(q) B_{\beta}(q) S_{\alpha\beta}(q). \quad (7)$$

Although the factorization of (5) is formally exact, it should be noted that, in contrast to the monodisperse case, the effective structure factor $S^{\text{eff}}(q)$ does not only depend on particle interactions (given by the partial structure factors) but also on the particle form amplitudes. Furthermore, the well-known relation between the isothermal compressibility and the structure factor at $q=0$ does not hold for $S^{\text{eff}}(q)$ (Hansen & McDonald, 1990).

Since the calculation of the partial structure factors from interaction potentials is usually quite involved, approximations for the effective structure factor are often used to fit scattering data. A simple form for $S^{\text{eff}}(q)$ would be an 'averaged structure factor'

$$S^{\text{ave}}(q) = \sum_{\alpha=1}^m x_{\alpha} S_{\alpha}(q) \quad (8)$$

where $S_\alpha(q)$ is the monodisperse structure factor for an equivalent system, consisting only of particles of species α . This means that (8) represents the weighted addition of partial structure factors $S_\alpha(q)$, each of them calculated for a monodisperse system with a fixed (total) volume fraction and varying radius. This averaged structure factor was exclusively used in our work introducing the GIFT method (Brunner-Popela & Glatter, 1997).

A similar expression has been introduced by Pedersen (1994) as the ‘local monodisperse approximation’

$$S^{\text{hma}}(q) = \frac{1}{P(q)} \sum_{\alpha=1}^m x_\alpha f_\alpha^2 B_\alpha^2(q) S_\alpha(q). \quad (9)$$

In contrast to the simple average, this approximation takes the dependence of the effective structure factor on the form factor into account, but it still neglects correlations between different species of particles, expressed by the cross terms in the matrix of the partial structure factor. In terms of interactions, each particle species interacts only with itself, which is an assumption that is reasonable for some applications (Mahalingam *et al.*, 1987; Li *et al.*, 1990).

Another frequently used approximation is the decoupling approximation (Pusey *et al.*, 1982; Kotlarchyk & Chen, 1983). In this case the effective structure factor for the polydisperse system can be derived from the structure factor $S(q)$ of a monodisperse system *via* the relation

$$S^{\text{dec}}(q) = R(q) + [1 - R(q)]S(q) \quad (10a)$$

with

$$R(q) = 1 - \left[\sum_{\alpha} x_\alpha f_\alpha B_\alpha(q) \right]^2 / \overline{P(q)}. \quad (10b)$$

If the particle interactions are the same for all species, this approximation becomes exact. The effect of the different scattering properties is then correctly accounted for by the factor $R(q)$. For large variations of the interaction potential, however, the representation of the particle correlations by a single monodisperse structure factor becomes inappropriate and therefore the decoupling approximation usually yields poor results.

The calculation of the partial and effective structure factors from model pair potentials $u_{\alpha\beta}$ is based on methods developed for the theory of liquids (Hansen & McDonald, 1990). Various review articles dealing with the application of these methods to suspensions of polydisperse colloidal particles are available (Salgi & Rajagopalan, 1993; D’Aguanno & Klein, 1996). For a given pair potential, accurate results for the structure factor can be obtained by computer simulation methods, but such simulations require too much computational time to be used in an inversion technique for scattering data like the GIFT. Therefore approximate results, based on the Ornstein–Zernike equation of the theory

of liquids, have to be used. The accuracy of these approximations can be tested by comparing them to computer simulations for typical interaction parameters.

In this work we consider hard-core potentials, which have been successfully applied to describe the structure of suspensions of nonionic colloidal particles. Structural and thermodynamic properties of a hard-sphere system do not depend on the temperature, but on the concentration and the size distribution. For this potential, the so-called mean spherical (MSA) or Percus–Yevick approximation (identical for hard spheres) provides an analytical solution for the monodisperse case (see *e.g.* Hansen & McDonald, 1990) as well as for mixtures (Blum & Stell, 1979; Vrij, 1979). The accuracy of this approximation has been demonstrated by Frenkel *et al.* (1986), who performed Monte Carlo simulations for polydisperse hard spheres and showed that the approximate results for the structure factor agree very well with the simulation data even for high concentrations and broad size distributions.

2.2.2. Non-spherical particles. So far only spherical particles have been considered. For non-spherical particles, the orientation dependence of the form amplitudes has to be taken into account, which implies that even in the monodisperse case, the scattering intensity can be only formally factorized as a form factor and a structure factor:

$$I(q) = NP(q)S^{\text{rod}}(q) \quad (11)$$

with

$$P(q) = \langle f^2(\Omega) B^2(q, \Omega) \rangle_\Omega. \quad (12)$$

If the non-spherical contributions to the pair potential can be neglected, the structure factor can be written as

$$S^{\text{rod}}(q) = 1 + \frac{\langle f(\Omega) B(q, \Omega) \rangle_\Omega^2}{P(q)} S(q) \quad (13)$$

where $S(q)$ is the structure factor for an equivalent system of monodisperse spherical particles and the angular brackets denote a simple orientational average

$$\langle \dots \rangle_\Omega = (1/4\pi) \int d\Omega. \quad (14)$$

Introducing the definition

$$R(q) = 1 - \frac{\langle f(\Omega) B(q, \Omega) \rangle_\Omega^2}{P(q)} \quad (15)$$

the expression for $S^{\text{rod}}(q)$ becomes identical to the decoupling approximation for the case of polydisperse spheres, (10). It should be noted that this analogy holds only as long as orientational correlations can be neglected.

A more exact evaluation of the scattering intensity from monodisperse non-spherical particles is considerably more difficult than in the spherical case, since it requires the calculation of both the spatial and the orientational correlations. Approximate results are

often obtained using a perturbation expansion around a spherical symmetric reference potential (Gray & Gubbins, 1984), whereas this approach gives only qualitatively correct results for suspensions of charged rod-like particles (Weyerich *et al.*, 1990; Canessa *et al.*, 1991). If the particles are also polydisperse, the problem becomes even more complicated. In the limiting case of slender cylindrical particles interacting *via* a hard-core potential, Shimada, Doi and Okano (Shimada *et al.*, 1988; Doi *et al.*, 1988) have proposed a method to calculate the structure factor, based on a mean-field theory, obtaining analytical results for the first terms of an expansion of the scattering vector. A numerical procedure to calculate the structure factor has been described by Maeda (1989). This method has been used to interpret neutron scattering data of micellar systems (Long *et al.*, 1994). Recently, van der Schoot & Odijk (1990) derived an analytical expression, which is almost identical to the numerical results of Maeda (1991). Furthermore, this derivation can be extended to the case of mixtures of particles with different lengths L_i (van der Schoot, 1992), leading to the formula for the effective structure factor

$$S^{\text{rod}}(q) = \frac{\overline{F(q)}}{\overline{F(q)} + 2c\overline{F(q)}^2 - (5/4)c\overline{G(q)}^2} \quad (16)$$

where c is a dimensionless quantity proportional to the number density. The functions F and G are given by

$$\overline{F(q)} = (\overline{L^2})^{-1} \sum_{i=1}^m x_i L_i^2 F(qL_i/2) \quad (17a)$$

with

$$F(x) = \frac{1}{x} \int_0^{2x} \frac{\sin x}{x} dx - \left(\frac{\sin x}{x} \right)^2 \quad (17b)$$

and

$$G(x) = \frac{3}{4} x^{-2} \left(1 - \frac{\sin 2x}{2x} \right) - \frac{1}{2} F(x) \quad (17c)$$

where $\overline{L^2}$ is the second moment of the length distribution. The function F represents the (normalized) form factor for infinitely thin rods (Berne & Pecora, 1976). Expression (16) is only valid for thin rods ($qR < 1$) with a large aspect ratio ($L/R \gg 1$).

2.3. Regularization of the structure factor

The GIFT method, as described in §2.1, allows for the choice of various structure factors for the evaluation of small-angle scattering data, for example, the averaged structure factor $S^{\text{ave}}(q)$ and the effective one $S^{\text{eff}}(q)$. If the latter is used, however, a serious problem arises: the solution of the nonlinear least-squares procedure for the determination of the coefficients of the structure factor is found by inversion of matrix $\mathbf{T}$ [see (3)] and, in general, such an inversion can be used only in the case of

basic functions that are not linearly dependent, which can be checked by the determinant of matrix $\mathbf{T}$. For normalized basic functions, the value of the determinant is one if the functions are orthogonal, and zero if they are linearly dependent. In the case of the averaged structure factor, $S^{\text{ave}}(q)$, the determinant is usually in the region of about 0.3 to 0.5 and the system is therefore far from being linearly dependent. The effective structure factor $S^{\text{eff}}(q)$, however, often has a determinant of $\mathbf{T}$ that is multiples of ten smaller than in the case of $S^{\text{ave}}(q)$. Thus, its parameters can be regarded to be quasi-linearly dependent. This quasi-linear dependence leads to numerical instability so that there is no chance to determine the global minimum of the parameter surface in an iterative procedure without further constraints. This ‘ill-posed problem’, however, can be regularized by a stabilization procedure:

$$\begin{aligned} \mathbf{t} &= (\mathbf{T} + \lambda_d \mathbf{K}) \Delta \mathbf{d} \\ \Delta \mathbf{d} &= (\mathbf{T} + \lambda_d \mathbf{K})^{-1} \mathbf{t} \\ \mathbf{K} &= \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix} \end{aligned} \quad (18)$$

where λ_d is a free stabilizing parameter and the matrix $\mathbf{K}$ is based on the minimization of the norm of the first derivative of the coefficients $\mathbf{d}$. This matrix has to be modified to a corresponding 4×4 matrix if charges are to be incorporated.

Regularization effects that the parameter increments $\Delta \mathbf{d}$ are no longer independent from each other. The degree of correlation is given by the magnitude of λ_d (Glatter, 1977a). We have tested a wide range of values for λ_d in a series of simulations [various parameter combinations and λ_d varying from 10^{-6} to 10^6 and normalized to the first element $\mathbf{T}(1, 1)$ of the matrix $\mathbf{T}$]. If the stabilization parameter is very low, no convergence can be achieved. If it is too high, however, the system is ‘frozen in’ because the parameters are correlated too strongly to each other. In between we found a broad range (many decades) of λ_d values giving a stable and converging solution, which indicates that this further regularization of the system does not falsify the solution of the least-squares problem. Thus, stabilization is not very sensitive to the actual value of λ_d and so we can use a fixed value of $\lambda_d = 10$, which is fairly in the middle of the allowed range, for all test cases. This avoids the variation of λ in a so-called λ -loop as it is necessary for the determination of the form-factor coefficients c_v .

3. Results

3.1. Structure factors for spherical particles

The model systems we consider in this work are particles with a monomodal size distribution character-

ized by a Schulz distribution (see *e.g.* Griffith, Triolo & Compere, 1987). We present results for the effective structure factor $S^{\text{eff}}(q)$ for polydisperse hard spheres, and we compare results for the average structure factor $S^{\text{ave}}(q)$, the local monodisperse approximation $S^{\text{hma}}(q)$ and the decoupling approximation $S^{\text{dec}}(q)$ with the full solution for the Percus–Yevick (PY) polydisperse structure factor $S^{\text{eff}}(q)$ using core-shell models for the intraparticle scattering properties. We assumed that the size polydispersity only affects the core radii, whereas the shell thickness is a constant. This is a reasonable assumption for microemulsion droplets with a single boundary layer of surfactant molecules.

For the calculation of the approximate effective structure factors, only the analytical expression of the PY solution for monodisperse hard spheres is needed. The effective structure factor for polydisperse hard spheres was calculated using Vrij’s solution of the PY closure for mixtures of hard spheres (Vrij, 1979). In Vrij’s solution, the scattered intensity of a discrete size distribution is represented by moments of the radii and the form amplitudes. The effective structure factor is derived from the scattered intensity by division through the average form factor. We do not reprint these expressions here because they can be found either in the original article or in the review paper of Salgi & Rajagopalan (1993). Instead of representing a polydisperse distribution of particle sizes by a discrete mixture, analytical results for a given continuous size distribution and form amplitudes can also be derived, which has been done in the case of homogeneous spheres with a Schulz size distribution by Griffith *et al.* (1987). This derivation, however, requires a considerable amount of algebraic manipulation, which would have to be redone for any new form of size distribution or form amplitude. Therefore, the representation by a discrete mixture seems to be a more flexible method if one is interested in a variety of possible intraparticle scattering models or size distributions. So a continuous size distribution of spheres can be represented by a discrete mixture if a sufficiently large number of species is used.

Fig. 2 shows the various structure factors for a polydisperse system of homogeneous spheres for a volume fraction Φ of 0.3 and a polydispersity μ of 30%. The averaged structure factor and the local monodisperse approximation both describe the influence of polydispersity by averaging over monodisperse structure factors. Since the structure factor of monodisperse hard spheres depends only on the volume fraction Φ and the product qR , the positions of the maxima and minima of the S_α are determined by the radii R_α . The averaging procedure leads then to a broadening of the structure peak and a smoothing of the oscillations in the structure factor. The local monodisperse approximation improves the simple averaged expression because it takes into account the dependence of $S(q)$ on the intraparticle scattering properties, using the form factors as weighting

factors in the averaging procedure. However, for $q = 0$, both approximations lead to the same incorrect result, which is identical to the value $S(0)$ for the structure factor of monodisperse spheres. The decoupling approximation is based on the opposite point of view: it represents the structure factor by a monodisperse $S(q)$ corrected by an ‘incoherent scattering’ term $R(q)$, which is typically most important for low q values. The decoupling approximation gives the worst results of all examined approximations, since it vastly overestimates the value of $S^{\text{dec}}(q)$ for small q values, whereas the broadening and damping of the structure peak are severely underestimated. Compared to the simple averaged structure factor $S^{\text{ave}}(q)$, the local monodisperse approximation $S^{\text{hma}}(q)$ provides much better accuracy. Apart from the region around the structure peak and of very small q values, the agreement with the PY solution is extremely good. The differences in the latter region look rather small on an absolute scale; however, as can be seen from the enlarged inset of Fig. 2, the relative difference between the local monodisperse and the PY value at $q = 0$ is about 30%.

In many cases, the effective structure factors for core-shell models have a similar form as for homogeneous or hollow spheres (see *e.g.* Robertus *et al.*, 1989). The situation changes considerably when we consider a model in which the scattering length of the shell is larger than the solvent scattering length, but the scattering length of the core is smaller. Depending on the ratio of the core and shell radii and the solvent scattering length, for a given particle size the total contrast will vanish. In a polydisperse suspension other particle sizes with non-vanishing contrast will be present, and the variation of form amplitudes will be more pronounced than in the case of homogeneous or hollow spheres. Since the effective structure factor for a similar model has only been examined for the special case $q = 0$ (Yan & Clarke,

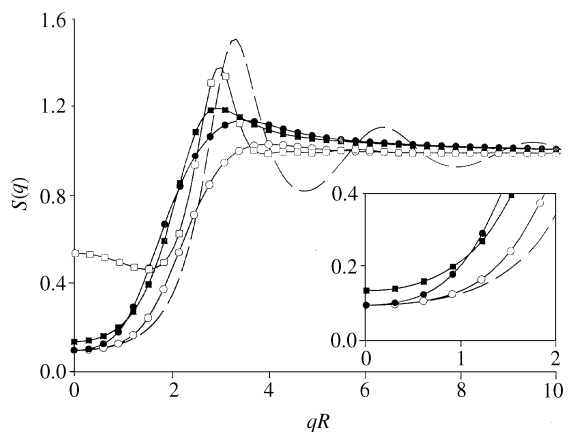


Fig. 2. Comparison of the effective structure factor $S^{\text{eff}}(q)$ for homogeneous spheres with different approximations: monodisperse $S(q)$ (dashed line), $S^{\text{ave}}(q)$ ($\circ$), $S^{\text{hma}}(q)$ ($\bullet$), $S^{\text{dec}}(q)$ ($\square$), PY $S^{\text{eff}}(q)$ ($\blacksquare$). $R = 10$ nm, $\Phi = 0.3$, $\mu = 30\%$.

1990), this case is discussed in more detail. The effective structure factor is calculated for different values of the polydispersity μ . The results are presented in Fig. 3 and demonstrate the sensitive dependence of $S^{\text{eff}}(q)$ on polydispersity. The increase of the effective structure factor with μ at small q values is about an order of magnitude larger than for homogeneous spheres. It should be noted, however, that this effect is based on the influence of the form amplitudes on the structure factor.

Finally, we compare the accuracy of approximation schemes for the effective structure factor. In Fig. 4, this comparison is made for a core-shell model with a mean radius of 10 nm and a fixed shell thickness of $d_{\text{shell}} = 3$ nm. The contrasts are chosen to be $\rho_{\text{shell}} - \rho_s = 1$ for the shell and $\rho_{\text{core}} - \rho_s = -1$ for the core. This system is quite close to the zero contrast that would be reached for $d_{\text{shell}} \simeq 2$ nm. The volume fraction and polydispersity are again set to $\Phi = 0.3$ and $\mu = 30\%$.

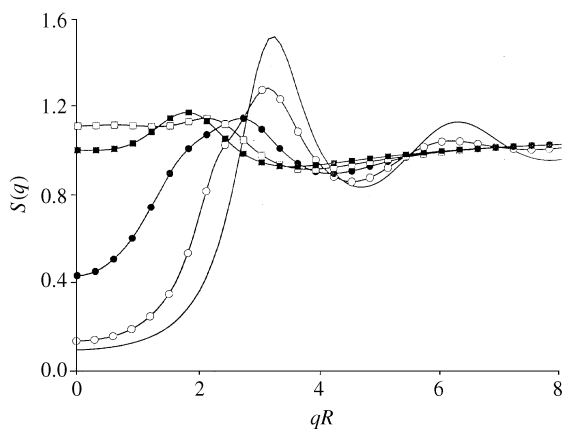


Fig. 3. Effective structure factor $S^{\text{eff}}(q)$ for core-shell spheres for different polydispersities: $\mu = 0\%$ (full line), $\mu = 10\%$ ($\circ$), $\mu = 20\%$ ($\bullet$), $\mu = 30\%$ ($\square$), $\mu = 40\%$ ($\blacksquare$). $\bar{R} = 10$ nm, $\Phi = 0.3$, $d_{\text{shell}} = 3$ nm, $\rho_{\text{shell}} - \rho_s = 1$, $\rho_{\text{core}} - \rho_s = -1$.

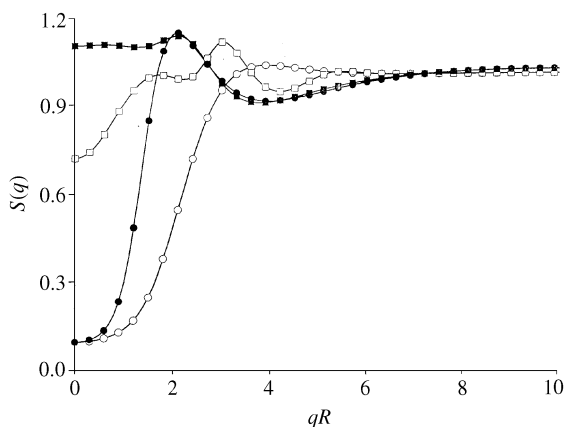


Fig. 4. Comparison of the effective structure factor $S^{\text{eff}}(q)$ for core-shell spheres with different approximations: $S^{\text{ave}}(q)$ ($\circ$), $S^{\text{lma}}(q)$ ($\bullet$), $S^{\text{dec}}(q)$ ($\square$), PY $S^{\text{eff}}(q)$ ($\blacksquare$). $\bar{R} = 10$ nm, $\Phi = 0.3$, $d_{\text{shell}} = 3$ nm, $\rho_{\text{shell}} - \rho_s = 1$, $\rho_{\text{core}} - \rho_s = -1$, $\mu = 30\%$.

Since the local monodisperse approximation does not include a dependence of $S^{\text{lma}}(0)$ on polydispersity, one might expect that for this case it will only provide a poor approximation. Surprisingly, the curves of the local monodisperse approximation and the PY solution already converge at the region of the structure peak and match closely for higher q values. The decoupling approximation, which is specially tailored for large variations in the scattering properties, provides only a qualitative description of the trends; the quantitative differences remain considerably large. For $q = 0$, the discrepancy can be as large as a factor of two, and the position of the structure peak is shifted to larger q values. The averaged structure factor is clearly a poor approximation; the interaction peak is completely smeared out at this high degree of polydispersity.

3.2. Structure factor for non-spherical particles

Last, but not least, we wish to present results for the structure factor of rod-like particles. The calculation of this quantity is, even for a monodisperse system, quite difficult if orientational correlations have to be taken into account. However, for the case of very slender cylindrical particles, a simple analytical expression based on a mean-field approximation can be derived [see equation (16)]. In Fig. 5, we present the results for monodisperse particles at three different volume fractions. The values for length and radius were chosen to be identical to the parameters reported in the publication by Long *et al.* (1994) for suspensions of mixed micelles. In this work the authors used a numerical evaluation of the mean-field theory for the interpretation of small-angle light and neutron scattering data. Their result for the effective structure factor is in good agreement with

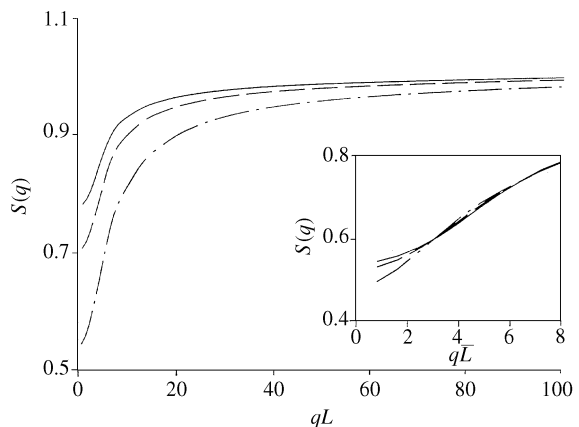


Fig. 5. Structure factor for monodisperse rod-like particles $S^{\text{rod}}(q)$ for different volume fractions: $\Phi = 0.0092$ (full line), $\Phi = 0.013$ (dashed line), $\Phi = 0.028$ (dashed-dotted line). $L = 80$ nm, $R = 2.7$ nm. Inset: structure factor for polydisperse rod-like particles $S^{\text{rod}}(q)$ for different polydispersities: $\mu = 0\%$ (monodisperse, full line), $\mu = 25\%$ (dashed line), $\mu = 50\%$ (dashed-dotted line). $L = 80$ nm, $R = 2.7$ nm, $\Phi = 0.028$.

Table . Parameters for the structure factor of a simulated scattering curve of homogeneous hard spheres and for its evaluation using either $S^{\text{ave}}(q)$ or $S^{\text{eff}}(q)$

	Φ	R_{HS}	μ	MD	Determinant of matrix $\mathbf{T}$
Simulation	0.15	10.0	0.4		
Averaged structure factor $S^{\text{ave}}(q)$	0.1357	13.6306	0.3185	1.0017	0.2193
Effective structure factor $S^{\text{eff}}(q)$	0.1504	10.0733	0.4013	0.9745	0.0010

the data in Fig. 5. Since the calculations are based on a mean-field theory, the structure factor is a monotonous increasing function of q and does not exhibit a correlation peak. This is certainly only a good approximation for sufficiently low volume fractions. But even for the small concentrations we used in our example, the value of $S^{\text{rod}}(0)$ deviates considerably from the non-interacting case $S^{\text{rod}}(q) = 1$. At this length of the cylinder and these volume fractions, the structure factor for hard spheres would be much closer to 1 for small scattering vectors. The expression for the effective structure factor includes the case for mixtures of cylinders with different lengths. Therefore it is possible to examine the influence of length polydispersity on $S^{\text{rod}}(q)$. We consider a discrete, monomodal distribution of lengths L_i with molar fractions x_i determined by a Schulz distribution characterized by the mean length $\bar{L}$ and polydispersity μ . The radius R is assumed to be constant for all species. Results for the largest volume fraction of Fig. 5 are presented in the inset. Even for a large polydispersity of 50%, the deviations from the monodisperse value are very small. Only at extremely small q values can a slight decrease of $S^{\text{rod}}(q)$ be observed. We conclude that length polydispersity of slender rod-like particles plays only a minor role for the effective structure factor. This statement is, however, restricted to monomodal size distributions. For binary mixtures a more pronounced dependence of the effective structure factor on the composition has been reported (van der Schoot, 1992).

In the discussion of the structure factors $S^{\text{rod}}(q)$ for different volume fractions, we want to stress the point that these functions do not show any interaction peak because they are based on the mean-field theory. This behavior can be modeled by $S^{\text{ave}}(q)$ with high enough polydispersity, which will be important in §3.5.

3.3. $S^{\text{eff}}(q)$ for homogeneous spheres and $S^{\text{ave}}(q)$

In the following we try to compare the utilization of the (easier but less exact) averaged structure factor $S^{\text{ave}}(q)$ and the (more complicated but more exact) effective structure factor $S^{\text{eff}}(q)$ in the inversion routine GIFT. So we present a simulated data set, very similar to an experimental one, that is evaluated using the generalized indirect Fourier transformation GIFT. We evaluate these scattering data using either the averaged or the effective structure factor to show that, within various limits, there is no large difference between them.

The form factor $P(q)$ is assumed to be a homogeneous sphere with a radius of 10 nm. The simulated effective structure factor $S^{\text{eff}}(q)$ has the following parameters: $\Phi = 0.15$, $R_{\text{HS}} = 10$ nm, $\mu = 0.4$. Smearing effects according to a typical small-angle neutron scattering experiment [wavelength distribution, full width at half-maximum (FWHM) = 15%] are taken into account. In addition, statistical noise is added. This simulated scattering intensity, $I(q)$, is evaluated for the determination of the form factor, represented in real space by the pair-distance-distribution function $p(r)$, and of the structure factor. On the one hand, this data set was evaluated using the effective structure factor $S^{\text{eff}}(q)$, and on the other hand, using the averaged structure factor $S^{\text{ave}}(q)$.

In Table 1, the corresponding parameter sets are listed. Using the effective structure factor $S^{\text{eff}}(q)$ allows for the exact recovery of the simulated data set (deviations less than 1%). The agreement of the theoretical function with the resulting one can be seen in Fig. 6. The determined $S^{\text{eff}}(q)$ is in excellent agreement with the simulated $S(q)$, and in regard to the pair-distance-distribution function $p(r)$, both curves coincide, too. This solution, however, can only be achieved by regularization of the structure factor. In Fig. 7, the development of the parameters (volume fraction Φ , radius R_{HS} , and polydispersity μ) during the iteration procedure is shown. Solving the normal equations without regularization (dashed lines) leads to pronounced instabilities

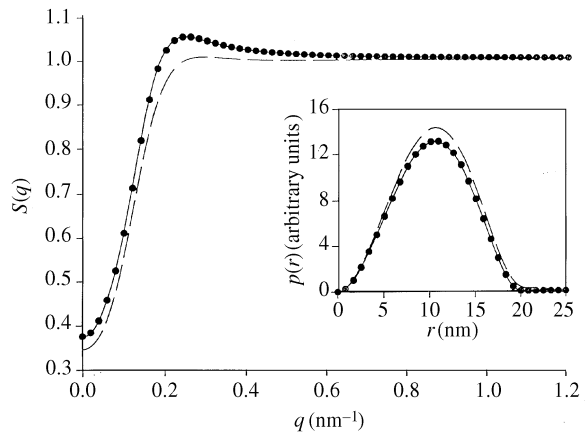


Fig. 6. Evaluation of simulated small-angle scattering data of homogeneous spheres [using $S^{\text{eff}}(q)$] with the GIFT under consideration of two different structure factors: simulated $S^{\text{eff}}(q)$ (●), resulting $S^{\text{eff}}(q)$ (full line), resulting $S^{\text{ave}}(q)$ (dashed line). Inset: corresponding pair-distance-distribution functions: simulated $p(r)$ (●), resulting $p(r)$ using $S^{\text{eff}}(q)$ (full line) or $S^{\text{ave}}(q)$ (dashed line).

in the development of the parameters so that the global minimum of the parameter surface cannot be found. Even the mean deviation does not converge to a value close to one. The corresponding determinant is determined to be 0.0010 (see Table 1), which means that the parameters are quasi-linearly dependent. But using a stabilization according to $\lambda_d = 10$ allows for a good convergence of the parameters to the excellent final values given in Table 1.

Regarding the solution of the averaged structure factor $S^{\text{ave}}(q)$, Table 1 shows quite a different parameter set. This behavior becomes clearer when we remember that the polydispersity is not correctly taken into account by $S^{\text{ave}}(q)$. This means that, while polydispersity of $S^{\text{eff}}(q)$ has a great influence on $S(0)$, the volume fraction is the only parameter of $S^{\text{ave}}(q)$ to vary $S(0)$. Thus, this quite different parameter set of $S^{\text{ave}}(q)$ leads to a similar result in the $p(r)$ function. Fig. 6 shows the determined structure factor to be significantly different from the simulated one, whereas no great difference in the $p(r)$ can be recognized. The difference remains more or less in the amplitude, so that normalization of the $p(r)$ makes the difference insignificant. The normalization factor can be determined as 0.92. The corresponding form factor $P(q)$ shows hardly any difference compared to the simulated one. The deviations are within graphical accuracy (results not shown), and a slight increase in forward intensity can be observed, which corresponds to a larger area of the $p(r)$ and to the too low value found for $S(0)$. As the determinant of matrix $\mathbf{T}$ for $S^{\text{ave}}(q)$ shows a relatively high value (0.219), no regularization has to be used in order to achieve optimum convergence.

Summing up, it is possible to evaluate scattering data using either the effective structure factor $S^{\text{eff}}(q)$ or the averaged structure factor $S^{\text{ave}}(q)$. In both cases the pair-distance-distribution function $p(r)$ can be recovered with a comparable mean deviation MD, and the form

factors $P(q)$ agree within graphical accuracy (results not shown). In the case of $S^{\text{ave}}(q)$ the solution of the inverse problem is quite easy, whereas in regard to $S^{\text{eff}}(q)$ we have to deal with an ill-posed problem requiring regularization.

3.4. $S^{\text{eff}}(q)$ for inhomogeneous spheres and $S^{\text{ave}}(q)$

As we have already mentioned, the GIFT should be a technique that can be used in as many cases as possible. It is based on a strict separation of the form factor and the structure factor, and as the former is absolutely model free, we prefer models for the structure factor that do not need information on the form amplitude either. From our experience we consider the averaged structure factor to be an easy but efficient approximation. Although its parameters are physically incorrect, the form factor can be recovered in the right way. In the previous section we applied $S^{\text{ave}}(q)$ to evaluate data which were simulated using $S^{\text{eff}}(q)$, and the result was satisfying.

Next we wish to perform a more complicated type of simulation for the case of inhomogeneous spheres. The whole procedure is based on a radial density distribution of a core-shell model, as shown in the inset of Fig. 10. The inner radius is 6 nm, the outer one 10 nm; the corresponding densities are -1.0 and $+1.0$, respectively. Including a polydispersity of 20%, the simulated form factor is calculated. The same profile is included in the calculation of the effective structure factor $S^{\text{eff}}(q)$ using the parameters $\Phi = 0.3$, $R_{\text{HS}} = 10$ nm, $\mu = 0.2$. Multiplication of $P(q)$ and $S(q)$, taking into account smearing effects according to a wavelength distribution (FWHM = 15%) and statistical noise, leads to simulated ‘experimental’ data (Fig. 8).

Now we try the simultaneous determination of the form factor and the structure factor from these data with

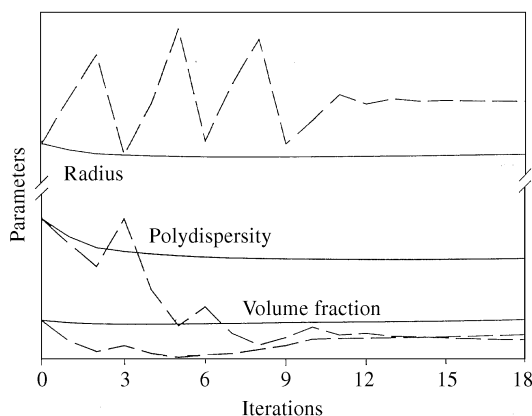


Fig. 7. Development of the volume fraction Φ , radius R_{HS} and polydispersity μ during the iteration procedure: optimum iteration including regularization (full lines), iteration without regularization (dashed lines).

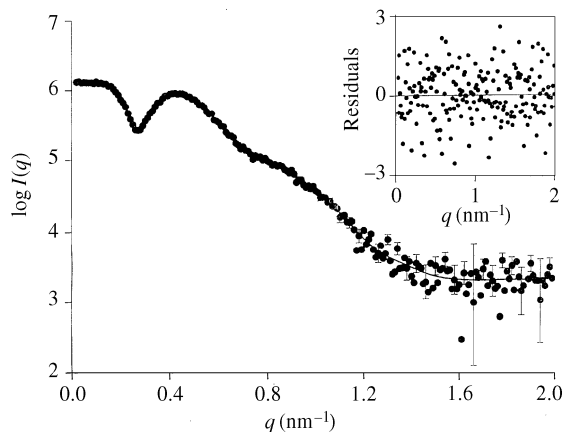


Fig. 8. Evaluation of simulated small-angle scattering data of inhomogeneous spheres with the GIFT under consideration of the effective structure factor for inhomogeneous spheres: simulated $I(q)$ (●), approximation function (full line). Inset: residuum plot.

an absolute minimum of *a priori* information. As, in principle, no information on a radial density profile is available in the beginning, we use the averaged structure factor $S^{\text{ave}}(q)$ in the first step because this type of $S(q)$ does not contain any information on the form amplitudes. Fig. 9 shows the simulated structure factor (dots) and the resulting averaged structure factor (dashed line). Although their shapes do not coincide completely, the shape of the pair-distance-distribution function can be recovered quite well, as can be seen in Fig. 10. Using the effective structure factor $S^{\text{eff}}(q)$ for homogeneous spheres in this first step does not lead to a satisfying result (dotted line). Both the structure factor (Fig. 9) and the pair-distance-distribution function (Fig. 10) show significant deviations from the simulated case.

The next step is to use the information about the form factor resulting from $S^{\text{ave}}(q)$ to find an improved $S^{\text{eff}}(q)$. As the $p(r)$ arising from the determination of $S^{\text{ave}}(q)$ was quite good, it was deconvoluted into a radial density profile under consideration of polydispersity. The resulting profile is shown in the inset of Fig. 10 (dashed line). There are slight deviations from the exact $\rho(r)$ used for the simulation, but this profile can be used in the calculation of the effective structure factor for inhomogeneous spheres, as described in the previous section, where size polydispersity only affects the core radii whereas the shell thickness is kept constant. The evaluation procedure is run again with this $S^{\text{eff}}(q)$, and the results are surprisingly good. The fit to the simulated data set is shown in Fig. 8 as well as the residuum plot. The $p(r)$ (Fig. 10) can be recovered with great accuracy, and the resulting structure factor (Fig. 9) coincides with the simulated one. The deconvolution of the $p(r)$ leads to a $\rho(r)$ being less exact (full line in the inset of Fig. 10); a second evaluation procedure using the new profile, however, improves the result significantly (dashed-

dotted line in the inset of Fig. 10). All corresponding parameter sets are listed in Table 2.

Regarding the results of this simulation, we want to stress the point that it is possible to evaluate scattering data without any information on the system. Therefore, in a first step the averaged structure factor is used to avoid any preliminary dependence on a model. The resulting $p(r)$ can be deconvoluted to obtain an approximate radial density profile that can be used afterwards to perform the whole evaluation procedure once again by means of the effective structure factor for inhomogeneous spheres. Thus, both the form factor and the structure factor can be recovered with great accuracy.

3.5. $S^{\text{rod}}(q)$, $S^{\text{eff}}(q)$ and $S^{\text{ave}}(q)$

Up to now we have restricted ourselves to spherical particles. The next step is to regard non-spherical systems and to check whether $S^{\text{ave}}(q)$ can be used in these cases, too.

The form factor is simulated according to the following parameters: diameter $\emptyset = 5$ nm, length $L = 25$ nm. The structure factor $S^{\text{rod}}(q)$ is calculated for $\Phi = 0.06$, $\emptyset = 5$ nm and length $L = 25$ nm. The corresponding scattering curve [product of $P(q)$ and $S^{\text{rod}}(q)$] is evaluated by the GIFT, either using the averaged structure factor $S^{\text{ave}}(q)$ or the effective structure factor $S^{\text{eff}}(q)$. The resulting parameters are listed in Table 3. As can be seen, the final parameters differ significantly from the simulated ones, which can be explained by the different structure-factor models used. Nevertheless, the

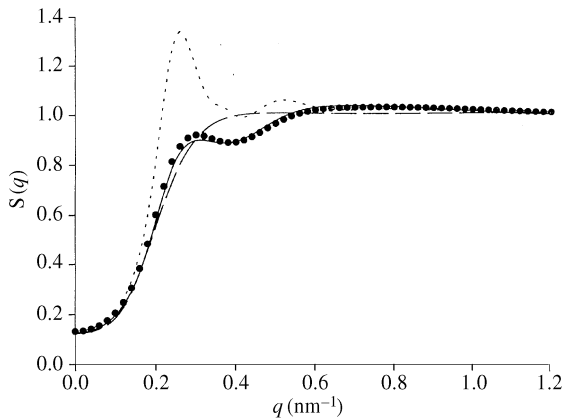


Fig. 9. Evaluation of simulated small-angle scattering data of inhomogeneous spheres [using $S^{\text{eff}}(q)$] with the GIFT under consideration of three different structure factors: simulated $S^{\text{eff}}(q)$ for inhomogeneous spheres (●), resulting $S^{\text{eff}}(q)$ for inhomogeneous spheres (full line), resulting $S^{\text{eff}}(q)$ for homogeneous spheres (dotted line), resulting $S^{\text{ave}}(q)$ (dashed line).

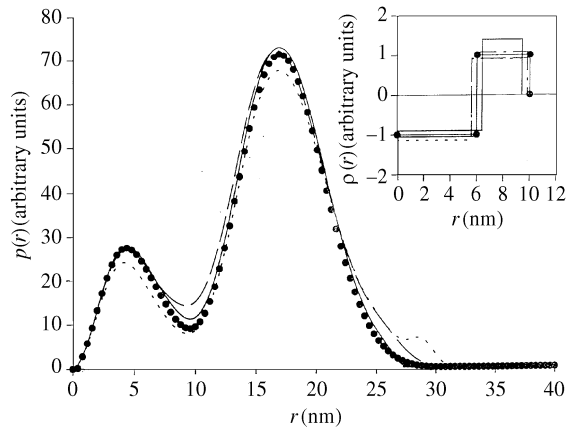


Fig. 10. Corresponding pair distance distribution functions arising from the evaluation of simulated small-angle scattering data of inhomogeneous spheres: simulated $p(r)$ for inhomogeneous spheres (●), resulting $p(r)$ using $S^{\text{eff}}(q)$ for inhomogeneous spheres (full line), resulting $p(r)$ using $S^{\text{eff}}(q)$ for homogeneous spheres (dotted line), resulting $p(r)$ using $S^{\text{ave}}(q)$ (dashed line). Inset: corresponding radial density profiles: simulated profile (●), profiles arising from the use of $S^{\text{eff}}(q)$ for inhomogeneous spheres in a first run (full line) or a second run (dashed-dotted line), profiles arising from the use of $S^{\text{eff}}(q)$ for homogeneous spheres (dotted line) or $S^{\text{ave}}(q)$ (dashed line).

Table . Parameters for the structure factor of a simulated scattering curve of inhomogeneous spherical particles and for its evaluation using either $S^{\text{ave}}(q)$ or $S^{\text{eff}}(q)$ for homogeneous or inhomogeneous spheres

	Φ	R_{HS}	μ	MD
Simulation	0.3	10.0	0.2	
Averaged structure factor $S^{\text{ave}}(q)$	0.2653	9.9713	0.3570	0.9861
Effective structure factor $S^{\text{eff}}(q)$ for homogeneous spheres	0.2630	11.7686	0.1068	1.0936
Effective structure factor $S^{\text{eff}}(q)$ for inhomogeneous spheres	0.3073	10.0017	0.2053	0.9727

Table . Parameters for the structure factor of a simulated scattering curve of rod-like particles and for its evaluation using either $S^{\text{eff}}(q)$ or $S^{\text{ave}}(q)$ for hard spheres

	Φ	R_{HS}	μ	MD
Simulation ($\emptyset = 5$ nm, $L = 25$ nm)	0.06	—	—	
Averaged structure factor $S^{\text{ave}}(q)$	0.037	5.5	0.92	1.0070
Effective structure factor $S^{\text{eff}}(q)$	0.050	2.0	0.80	0.9964

result in regard to the pair-distance distribution functions is quite good, which is demonstrated in the inset of Fig. 11. The $p(r)$ can always be recovered to be of the shape of a cylindrical particle. The result using $S^{\text{ave}}(q)$ is better than the $p(r)$ using $S^{\text{eff}}(q)$. Regarding the structure factors themselves, Fig. 11 indicates that the averaged structure factor for spheres is able to model the shape of the structure factor for rod-like particles much better than the effective structure factor. This can easily be understood by comparing the structure factors in Figs. 2 and 5. $S^{\text{rod}}(q)$ increases monotonically from $S(0)$ to its plateau value of one. A similar behavior is found for higher polydispersity for $S^{\text{ave}}(q)$, whereas $S^{\text{eff}}(q)$ always shows an interaction peak with a value higher than one. The parameters for $S^{\text{ave}}(q)$, of course, are far from the correct values, which can be seen in the very high values for the polydispersity. It must be kept in

mind that replacing $S^{\text{rod}}(q)$ by $S^{\text{ave}}(q)$ is just a qualitative analogy.

Summing up, it is possible to evaluate scattering data of rod-like particles using other models for the structure factor than the exact $S^{\text{rod}}(q)$. The averaged structure factor $S^{\text{ave}}(q)$ especially enables us to recover the form factor rather well.

4. Discussion

The generalized indirect Fourier transformation (GIFT), introduced in the first paper of this series, is a new technique for the evaluation of small-angle X-ray and neutron scattering data of solutions at higher concentrations. It is based on the simultaneous determination of the form factor and the structure factor with a minimum of *a priori* information. The former is absolutely model free, whereas the latter has to be calculated according to an interaction potential. In the former publication, we restricted ourselves to a very simple model for the interparticle contributions that we call the averaged structure factor. This paper, however, introduces other types of structure factors. On the one hand, we have presented the effective structure factor for polydisperse hard spheres that is calculated by the Percus–Yevick approximation and compared it to various approximations, such as the averaged structure factor, the local monodisperse approximation and the decoupling approximation. On the other hand, we have presented results for the effective structure factor for polydisperse rod-like particles calculated by the mean-field approximation.

One essential point in regard to the definition of the structure factors is the factorization of the scattering intensity that is necessary for the determination of the form factor under consideration of the structure factor. Thus, the effective structure factor for polydisperse systems does not only depend on particle interactions (given by the partial structure factors), but also on the particle form amplitudes. We have calculated the structure factor for homogeneous and for core-shell particles, and some results were quite surprising.

In the case of homogeneous systems the results of the local monodisperse approximation agree very well with the PY results, despite the fact that this type of approximation does not describe the dependence of the effective structure factor on polydispersity at $q = 0$. This leads to significant discrepancies for homogeneous

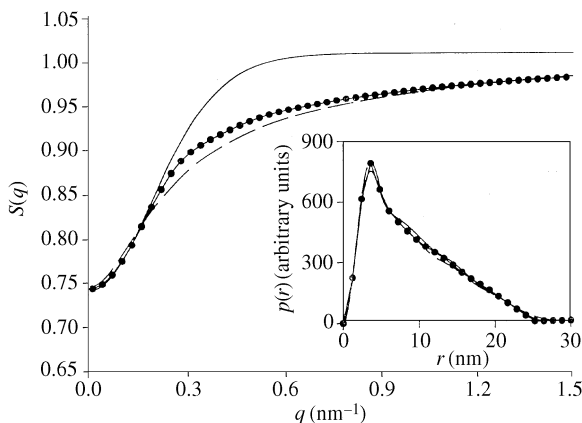


Fig. 11. Evaluation of simulated small-angle scattering data of homogeneous rod-like particles [using $S^{\text{rod}}(q)$] with the GIFT under consideration of two different structure factors for spheres: simulated $S^{\text{rod}}(q)$ (●), resulting $S^{\text{eff}}(q)$ (full line), resulting $S^{\text{ave}}(q)$ (dashed line). Inset: corresponding pair-distance-distribution functions: simulated $p(r)$ (●), resulting $p(r)$ using $S^{\text{eff}}(q)$ (full line) or $S^{\text{ave}}(q)$ (dashed line).

spheres at very low q values. The decoupling approximation gives only poor results, overestimating the value of the effective structure factor at small scattering vectors and in the region of the structure peak.

For a core-shell model which is characterized by a different sign of the scattering contrast for the core region to that of the shell region, the variation of the shell thickness at constant polydispersity changes the effective structure factor drastically, although the particle interactions are not affected by this variation. The dependence on polydispersity at low q values for a core-shell model with low mean scattering contrast was an order of magnitude larger than in the case of homogeneous spheres. In this case the local monodisperse approximation still agrees well with the PY results for sufficiently large q values, although for principal reasons it cannot describe the behavior at small scattering vectors. The decoupling approximation fails also in this case to describe the effects of polydispersity appropriately. It only reproduces the variation of the effective structure factor at small scattering vectors qualitatively; the quantitative differences, however, remain large.

In both cases, the averaged structure factor yields only rather poor results. Nevertheless, there exist serious arguments in favor of using this structure factor in a first evaluation procedure, which will be explained below. The decoupling approximation is unreliable for describing the effects of polydispersity in systems of hard spheres, irrespective of the intraparticle scattering properties. The local monodisperse approximation, however, can be a valuable substitute for the more complex PY expressions, at least for particles for which the size dependence of the particle scattering properties is not too large.

If it is possible to change the scattering contrast of the core and shell region in experiments (e.g. in neutron scattering), the influence of the particle scattering properties on the effective structure factor could be exploited. By avoiding situations with low mean contrast of the scatterers, it is possible to reduce the influence of the polydispersity on the structure factor. For systems with sufficiently high polydispersity, however, the influence of particle interactions on the scattering curve can be diminished by preparing particles with low mean contrast, since in this case the value of the effective structure factor is close to one for all scattering vectors, even at high volume fractions.

Besides systems of spherical particles, we have investigated the influence of length polydispersity on the structure factor of slender rod-like particles, using a simple mean-field theory for the calculation of the structure factor. In contrast to spheres, this influence is very small for a monomodal size distribution, at least at this level of approximation.

Our aim in the evaluation of small-angle scattering data is a model-free determination of the form factor or

its Fourier transform, the pair-distance-distribution function. Therefore, to our minds it is more convenient to use the averaged structure factor in a first evaluation procedure for the following reasons. From experimental results we know that typical hard spheres are hardly ever found. Especially in the case of microemulsions, for example, we have to deal with shape and size polydispersity so that the assumption of hard spheres cannot be correct. The pronounced peak of the effective structure factor, however, appears for hard-sphere interactions in the case of uncharged systems. Microemulsions with size and shape fluctuations ('soft' spheres) do not show such a pronounced peak but only a reduction at low q values. This behavior, however, can easily be described by the averaged structure factor. Furthermore, this type of structure factor does not contain any information on particle form amplitudes, so that direct structure analysis without *a priori* information is possible. We have shown that the averaged structure factor is able to replace both the effective structure factor of spherical (homogeneous and inhomogeneous) systems and the structure factor for rod-like particles, although the parameters are not correct and sometimes deviate considerably from the correct values. The deviation of the structure factor at low q values only leads to a slightly wrong amplitude in the form factor (in real and reciprocal space) but does not change its functional form. Finally, the evaluation procedure in the GIFT is much easier in the case of the averaged structure factor because no extra regularization is needed, which is the case, however, if we use the effective structure factor. Although we were able to solve this instability in the convergence of the nonlinear least-squares procedure by means of introducing a stabilization of the structure-factor parameters, the problem is much more complicated. We do not see any advantage in using the decoupling approximation as it includes particle form factors in the definition of the model of the structure factor. The same holds for the local monodisperse approximation. Anyway, if we allow for such a complicated model, then we prefer the best one available, namely the effective structure factor.

What we suggest then, is to use the averaged structure factor in the first step in order to determine the pair-distance-distribution function. In a following step, this information can be used to replace this simple model for the structure factor by a more correct one. This means, if the particles are determined to be spherical, we can use the effective structure factor for spherical systems, but if they are elongated, the structure factor for rod-like particles is adequate. In this way, correct parameters for the interparticle effects can be determined.

This work was supported by the 'Österreichischer Fonds zur Förderung der wissenschaftlichen Forschung' under Grants P10199-CHE, P10682-CHE and P12611-CHE.

References

- Ailawadi, N. K. (1980). *Phys. Rep.* **57**, 241–306.
- Berne, J. B. & Pecora, R. (1976). *Dynamic Light Scattering*. New York: John Wiley.
- Blum, L. & Stell, G. (1979). *J. Chem. Phys.* **71**, 42–46; erratum (1980), **72**, 2212.
- Brumberger, H. (1995). *Modern Aspects of Small-Angle Scattering*. Dordrecht: Kluwer.
- Brunner-Popela, J. & Glatter, O. (1997). *J. Appl. Cryst.* **30**, 431–442.
- Canessa, E., D'Aguanno, B., Weyerich, B. & Klein, R. (1991). *Mol. Phys.* **73**, 175–197.
- D'Aguanno, B. & Klein, R. (1996). *Static Properties of Colloidal Suspensions*, in *Light Scattering. Principles and Development*, edited by W. Brown. Oxford: Clarendon Press.
- Doi, M., Shimada, T. & Okano, K. (1988). *J. Chem. Phys.* **88**, 4070–4075.
- Feigin, L. A. & Svergun, G. I. (1987). *Structure Analysis by Small-Angle X-ray and Neutron Scattering*, edited by G. W. Taylor. New York: Plenum Press.
- Frenkel, D., Vos, R. J., de Kruif, C. G. & Vrij, A. (1986). *J. Chem. Phys.* **84**, 4625–4630.
- Glatter, O. (1977a). *Acta Phys. Austr.* **47**, 83–102.
- Glatter, O. (1977b). *J. Appl. Cryst.* **10**, 415–421.
- Glatter, O. (1979). *J. Appl. Cryst.* **12**, 166–175.
- Glatter, O. (1980a). *J. Appl. Cryst.* **13**, 7–11.
- Glatter, O. (1980b). *J. Appl. Cryst.* **13**, 577–584.
- Glatter, O. & Kratky, O. (1982). *Small-Angle X-ray Scattering*. London: Academic Press.
- Gray, C. G. & Gubbins, K. E. (1984). *Theory of Molecular Liquids*. Oxford: Clarendon Press.
- Griffith, W. L., Triolo, R. & Compere, A. L. (1987). *Phys. Rev. A*, **35**, 2200–2206.
- Guinier, A. & Fournet, G. (1955). *Small-Angle Scattering of X-rays*. New York: John Wiley.
- Hansen, J. P. & McDonald, I. R. (1990). *Theory of Simple Liquids*. London: Academic Press.
- Hayter, J. B. (1985). In *Physics of Amphiphiles: Micelles, Vesicles and Microemulsions*, edited by V. Degiorgio. Amsterdam: North-Holland.
- Hayter, J. B. & Penfold, J. (1981). *Mol. Phys.* **42**, 109–118.
- Kaler, E. W. (1988). *J. Appl. Cryst.* **21**, 729–736.
- Kaler, E. W. (1995). In *Modern Aspects of Small-Angle Scattering*, edited by H. Brumberger. Kluwer Academic Publishers.
- Kotlarchyk, M. & Chen, S.-H. (1983). *J. Chem. Phys.* **79**, 2461–2469.
- Li, Q., Kesternich, W., Schroeder, H., Schwahn, D. & Ullmaier, H. (1990). *Acta Metall. Mater.* **38**, 2383–2392.
- Long, M. A., Kaler, E. W., Lee, S. P. & Wignall, G. D. (1994). *J. Phys. Chem.* **98**, 4402–4410.
- Maeda, T. (1989). *Macromolecules*, **22**, 1881–1890.
- Maeda, T. (1991). *Macromolecules*, **24**, 2740–2747.
- Mahalingam, K., Gu, B. P., Liedl, G. L. & Sanders, T. M. Jr (1987). *Acta Metall.* **35**, 483–498.
- Pedersen, J. S. (1994). *J. Appl. Cryst.* **27**, 595–608.
- Percus, J. K. (1962). *Phys. Rev. Lett.* **8**, 462–463.
- Percus, J. K. & Yevick, G. J. (1958). *Phys. Rev.* **110**, 1–13.
- Pusey, P. N., Fijnaut, H. M. & Vrij, A. (1982). *J. Chem. Phys.* **77**, 4270–4281.
- Robertus, C., Philipse, W. H., Joosten, J. G. H. & Levine, Y. K. (1989). *J. Chem. Phys.* **90**, 4482–4490.
- Salgi, P. & Rajagopalan, R. (1993). *Adv. Coll. Int. Sci.* **43**, 169–288.
- Schoot, P. van der (1992). *Macromolecules*, **25**, 2923–2927.
- Schoot, P. van der & Odijk, T. (1990). *Macromolecules*, **23**, 4181–4182.
- Shimada, T., Doi, M. & Okano, K. (1988). *J. Chem. Phys.* **88**, 2815–2821.
- Vrij, A. (1979). *J. Chem. Phys.* **71**, 3267–3270.
- Weyerich, B., D'Aguanno, B., Canessa, E. & Klein, R. (1990). *Faraday Discuss. Chem. Soc.* **90**, 245–259.
- Yan, Y. D. & Clarke, J. H. R. (1990). *J. Chem. Phys.* **93**, 4501–4509.